



**Bayesian inference and variant identification in
metagenomic data:**

Subsampling methods for massive datasets

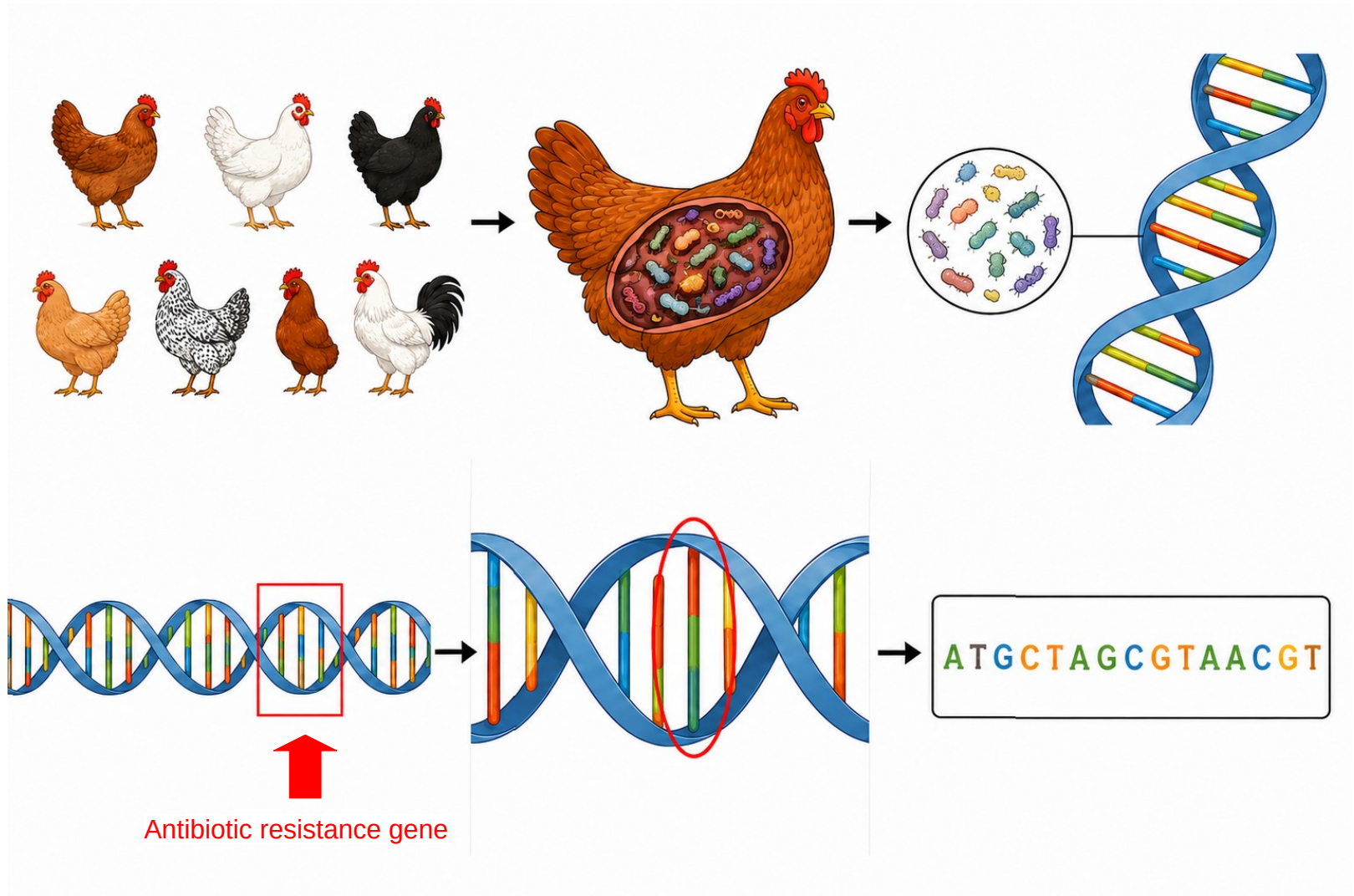
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1 September , 2026

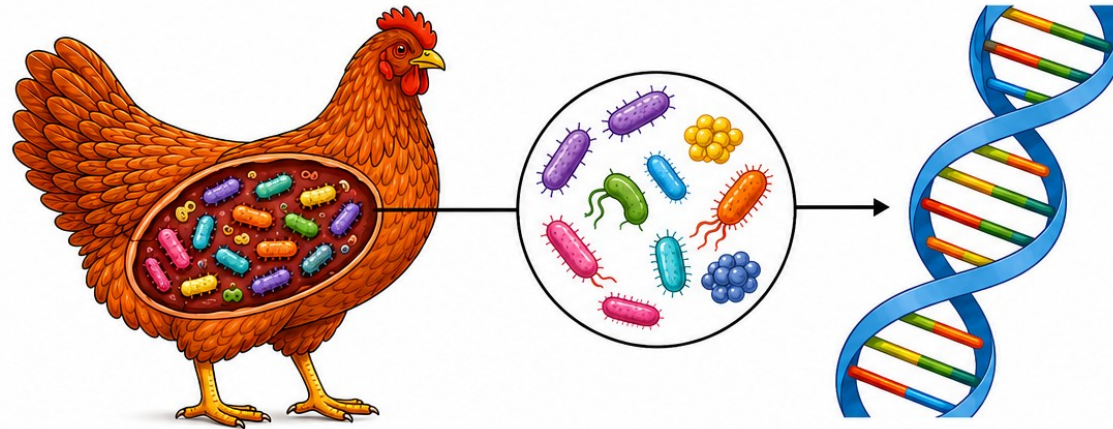


Supervisors: Anne Laure Abraham, Daniel Bonnery,
Guillaume Kon Kam King,

What are we looking at in this study?



Variant profiling and read extraction



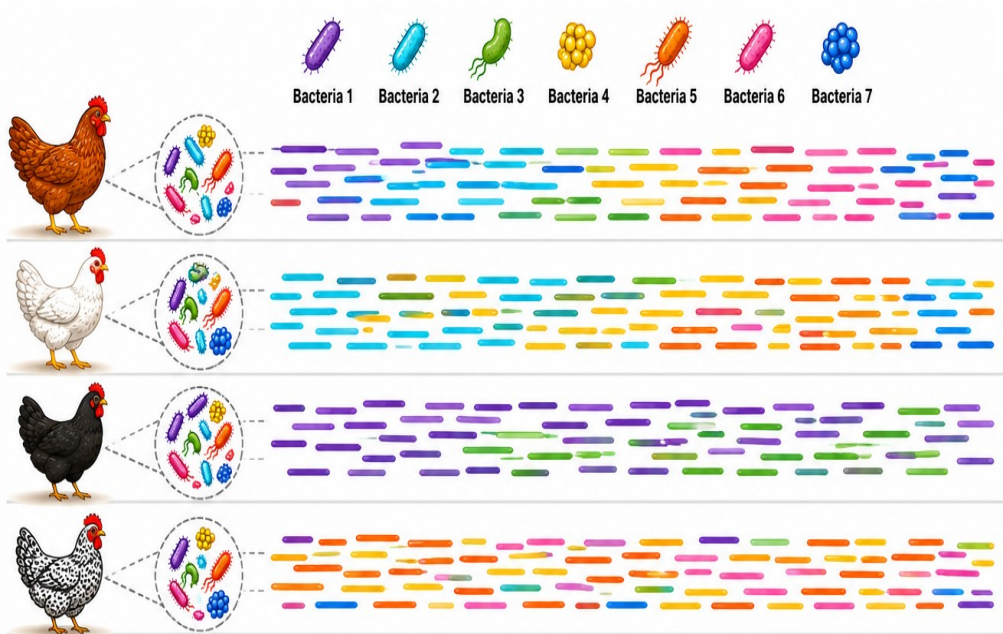
COMPLETE SEQUENCE GENOME

ATGCCGATCGTACGCTAGCTAGCTA GCTAGCTAGCTAACTGATCGATGCCA GATCCGATCGGATATCGATCGATCGA GCTATCGCTAGCTCTGATCGGATCGA GCTAGCTAACGATCCGATCGATTAC CGATCGATCCGATTACGATCGATGCTA

READS FROM THE GENOME

ATGCCGATC	GCTAGCTAG	CTGATCGATG	GATCCGATCG	ATCGATCGAT	GCTAGCTAAC	CGATCGATCC
GCCGATCGTA	CTAGCTAACT	TGATCGATCG	ATCCGATCGG	TCGATCGATC	CTAGCTAACG	GATCGATCCG
TCGATCGTAC	AGCTAACTG	GATCGATGCC	CCGATCGGAT	CGATCGATCG	TAGCTAATTA	ATCGATCCGA
CGTACGCTAG	CTAACTGAT	ATCGATGCCA	CCGATATCGA	GATCGATCGA	CTAATACGAT	CCGATTACGA
ACGCTAGCTA	ACTGATCGAT	TCGATGCCAT	ATATCGATCG	ATCGATCGGA	ACGATCCGAT	GATTACGATC

Problem Formulation and Objectives



Observations

S : Number of samples

$\mathcal{R}$: Set of reads

$X_{r,v,a}$: 1 if read r at position v has nucleotide a ,
0 otherwise

Model Parameters

G : Number of variants

$\tau_{g,v,b}$: 1 if variant g has nucleotide b at position
 v

$\pi_{g,s}$: Relative abundance of variant g in sample s

ε : Sequencing error rate

Objective	Description
1	Reconstruction of the latent sequences of the variants τ present in the data.
2	Estimation of the latent relative abundances $\pi_{.,s}$ of each variant in each individual.

Mixture model

Read likelihood per position given $G = 1$:

$$\mathbb{P}(X_{r,v,a} = 1 \mid \tau_g, \epsilon) = \sum_{b \in \mathcal{A}} \tau_{g,v,b} \epsilon_{b,a}.$$

Sequencing
noise

Likelihood of the full read given $G = 1$:

$$\mathbb{P}(X_r \mid \tau_g, \epsilon) = \prod_{v \in \mathbf{v}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \epsilon_{b,a} \right)^{X_{r,v,a}}.$$

Product by
positions

Mixture model likelihood of the read under $G > 1$:

$$\mathbb{P}(X_r \mid \tau, \epsilon, \pi_{\cdot, s_r}) = \sum_{g=1}^G \pi_{g, s_r} \mathbb{P}(X_r \mid \tau_g, \epsilon).$$

Weighted sum
by relative
abundances for
each variant

Full data likelihood under a mixture model with $G > 1$:

$$\mathbb{P}(X \mid \tau, \epsilon, \pi) = \prod_{r \in \mathcal{R}} \sum_{g=1}^G \pi_{g, s_r} \prod_{v \in \mathbf{v}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \epsilon_{b,a} \right)^{X_{r,v,a}}.$$

Assuming
independence of
reads, likelihood
approach of the
model

Product over many terms

Mixture model
over G

Posterior sampling for mixture models

Bayesian formula :

$$p(Y \mid X) \propto p(Y) p(X \mid Y)$$

Prior Distribution

$$\pi_{\cdot,s} \sim \text{Dirichlet}(\alpha)$$

(Z : Latent allocation variable)

$$Z_r \mid \pi_{\cdot,s_r} \sim \text{Categorical}(\pi_{\cdot,s_r})$$

$$\varepsilon \sim \text{Beta}(a, b)$$

$$\tau_{g,v} \sim \text{Categorical}(\theta_{g,v,1:4})$$

Conditional Posterior distribution

Bayes
 $\Rightarrow$

$$\pi_{\cdot,s} \mid Z, \varepsilon, X, \tau \sim \text{Dirichlet}(\alpha + n_{1,s}, \dots, \alpha + n_{G,s})$$

Bayes
 $\Rightarrow$

$$Z_r \mid X_r, Z_{-r}, \tau, \varepsilon, \pi \sim \text{Categorical} \left(\frac{\pi_{g,s_r} p(X_r \mid \varepsilon, \tau_g)}{\sum_{g' \in \{1, \dots, G\}} \pi_{g',s_r} p(X_r \mid \varepsilon, \tau_{g'})} \right)_{g=1}^G$$

Bayes
 $\Rightarrow$

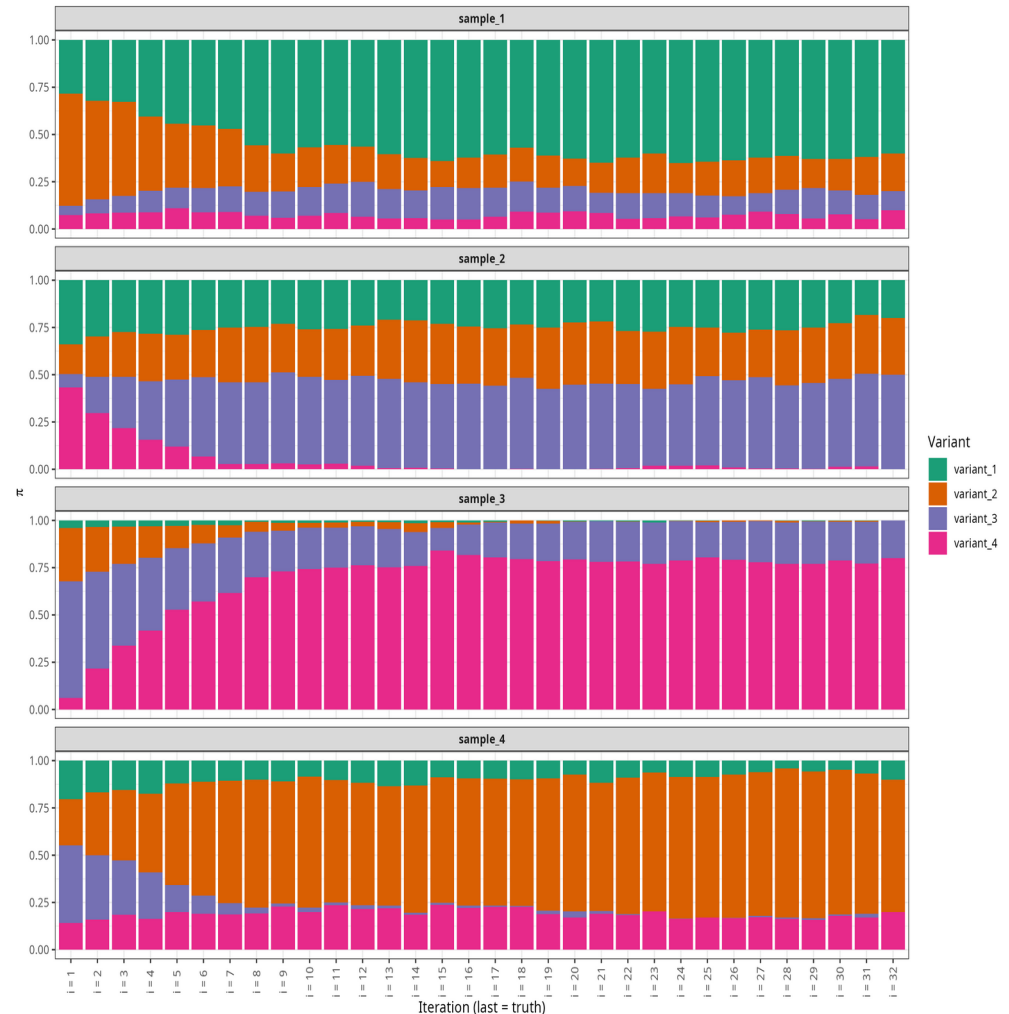
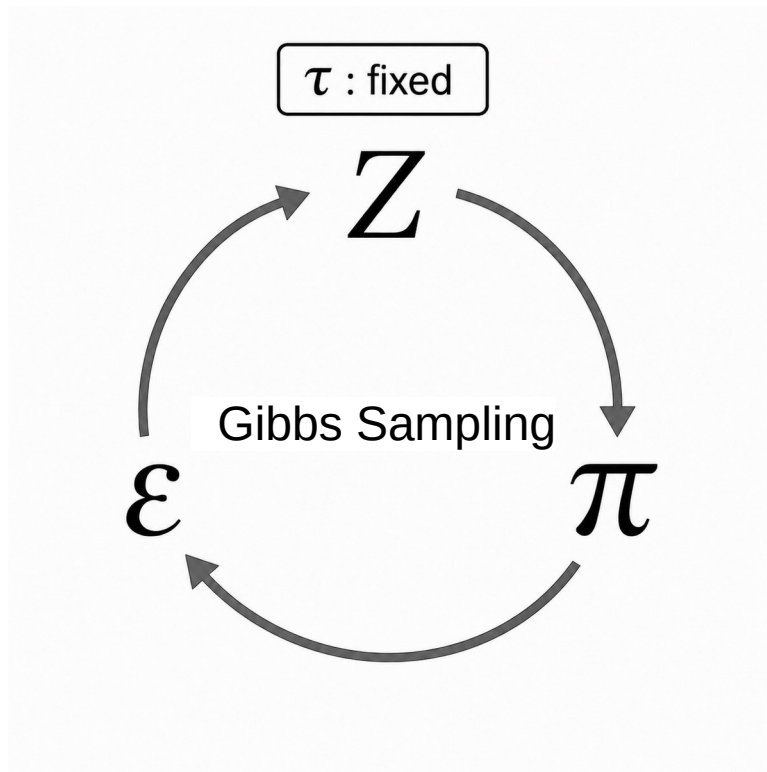
$$\varepsilon \mid X, Z, \pi, \tau \sim \text{Beta}(a + E, b + M)$$

Bayes
 $\Rightarrow$

$$\tau_{g,v} \mid X, Z, \varepsilon, \pi \sim \text{Categorical} \left(\frac{\theta_{g,v,b} (1 - \varepsilon)^{m_{g,v,b}} \left(\frac{\varepsilon}{3}\right)^{n_{g,v} - m_{g,v,b}}}{\sum_{b' \in \mathcal{A}} \theta_{g,v,b'} (1 - \varepsilon)^{m_{g,v,b'}} \left(\frac{\varepsilon}{3}\right)^{n_{g,v} - m_{g,v,b'}}} \right)_{b \in \mathcal{A}}$$

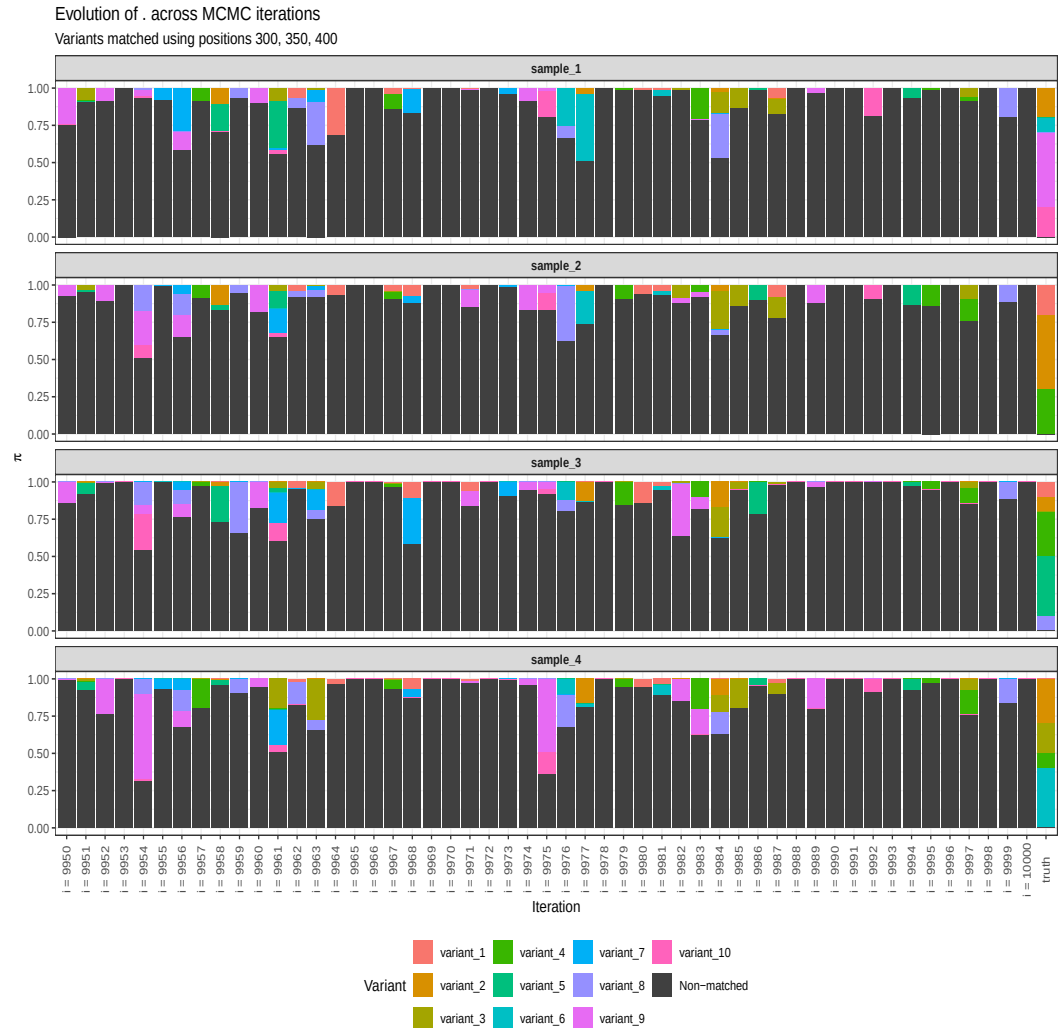
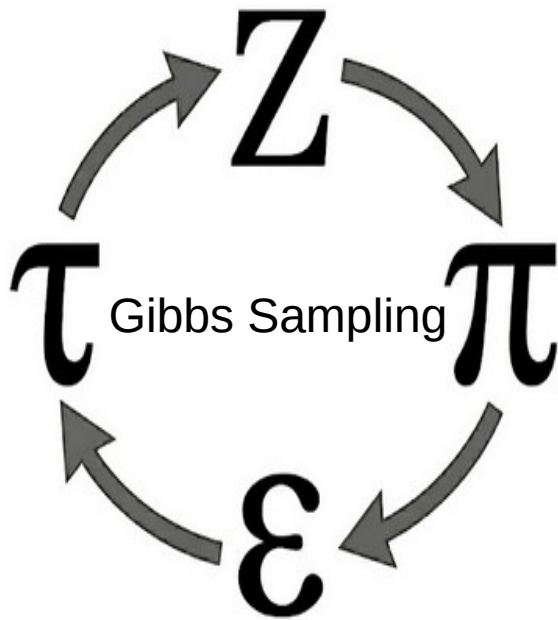
Markov chain and Monte Carlo algorithms *(Known-variant setting)*

Fast convergence to the stationary distribution



Markov chain and Monte Carlo algorithms (Unknown-variant setting)

Bad mixing Markov chains



Likelihood approximation methods

Global Objective :

- Computational cost
- Approximation quality
- Performance vs complexity reduction

Sub-sampling strategies:

-Read Subsampling:

- Random
- Coverage

-Position Subsampling

-Mixed strategies

Computational optimization:

For the likelihood, a read is characterized by its sample of origin, the positions it covers, and the observed bases.

We define equivalence classes of reads that share the same likelihood value in order to avoid redundant computations.

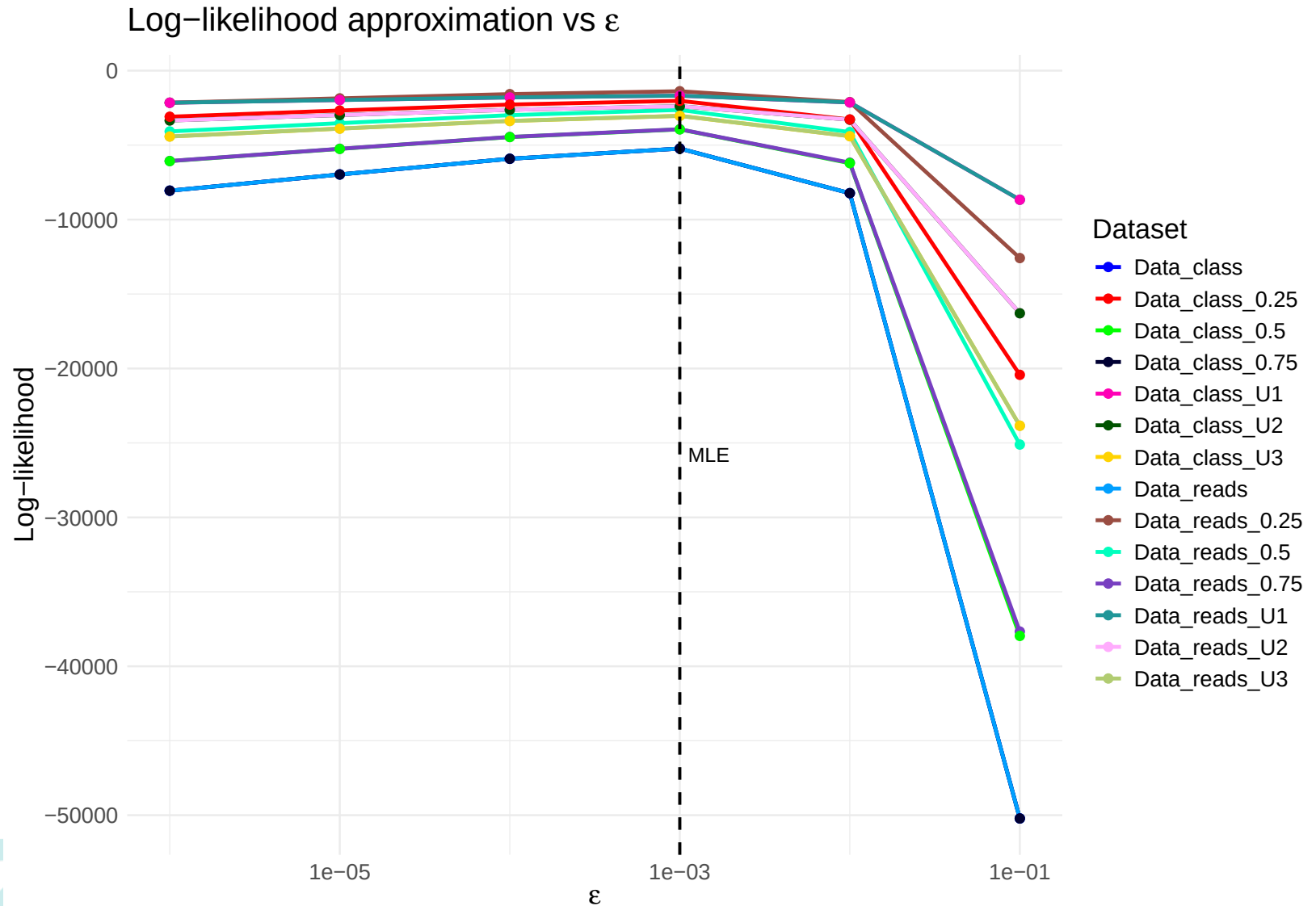
$$\boxed{\begin{array}{l} r_i \sim r_j \iff \begin{array}{l} s_i = s_j \\ P_i = P_j \\ B_i = B_j \end{array} \end{array}}$$

$$\tilde{r} = [r] = \{ r' \in \mathcal{R} : r' \sim r \}$$

Detailed Description of the Selected Methods

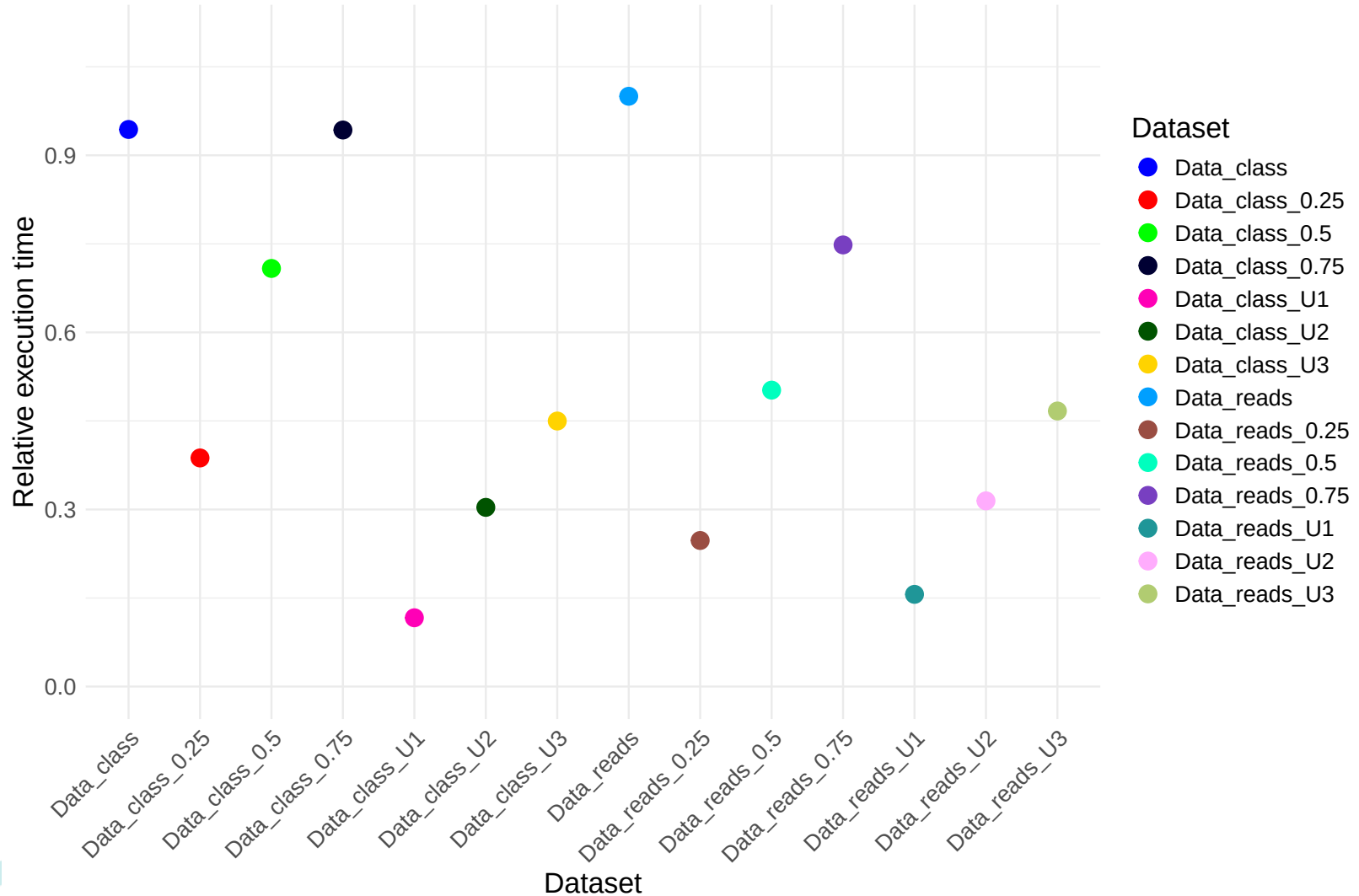
Name	Method	U/%	Computational optimization
Data_reads	All Data	-	No
Data_reads_U1	Position subsampling	U1 = [300,400]	No
Data_reads_U2	Position subsampling	U2 = [250,450]	No
Data_reads_U3	Position subsampling	U3 = [200,500]	No
Data_reads_0.25	Coverage subsampling	25%	No
Data_reads_0.5	Coverage subsampling	50%	No
Data_reads_0.75	Coverage subsampling	75%	No
Data_class	All Data	-	Yes
Data_class_U1	Position subsampling	U1 = [300,400]	Yes
Data_class_U2	Position subsampling	U2 = [250,450]	Yes
Data_class_U3	Position subsampling	U3 = [200,500]	Yes
Data_class_0.25	Coverage subsampling	25%	Yes
Data_class_0.5	Coverage subsampling	50%	Yes
Data_class_0.75	Coverage subsampling	75%	Yes

Benchmark: Likelihood Approximations



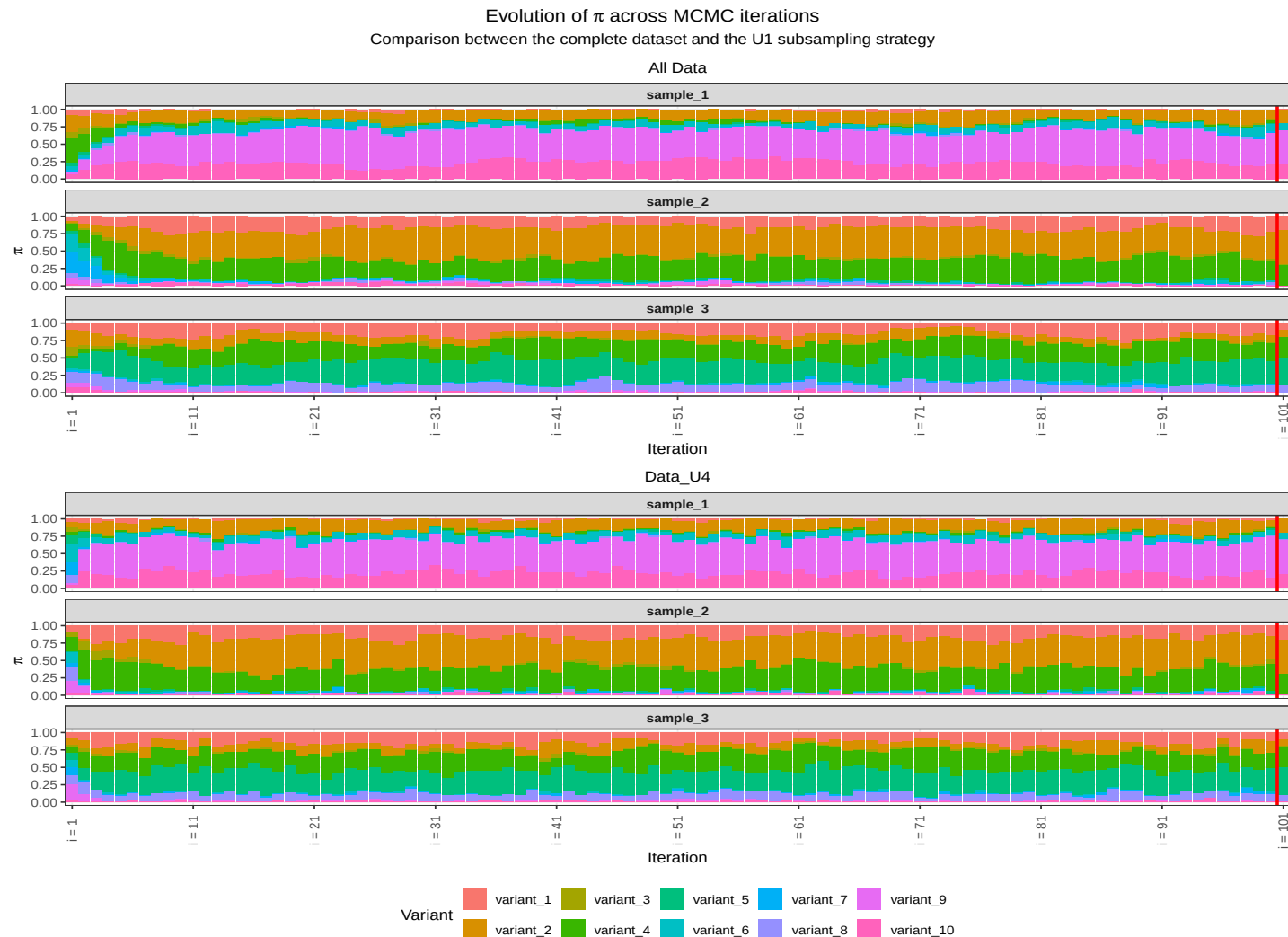
Benchmark: Execution Time Comparison

Relative execution time with respect to the slowest dataset



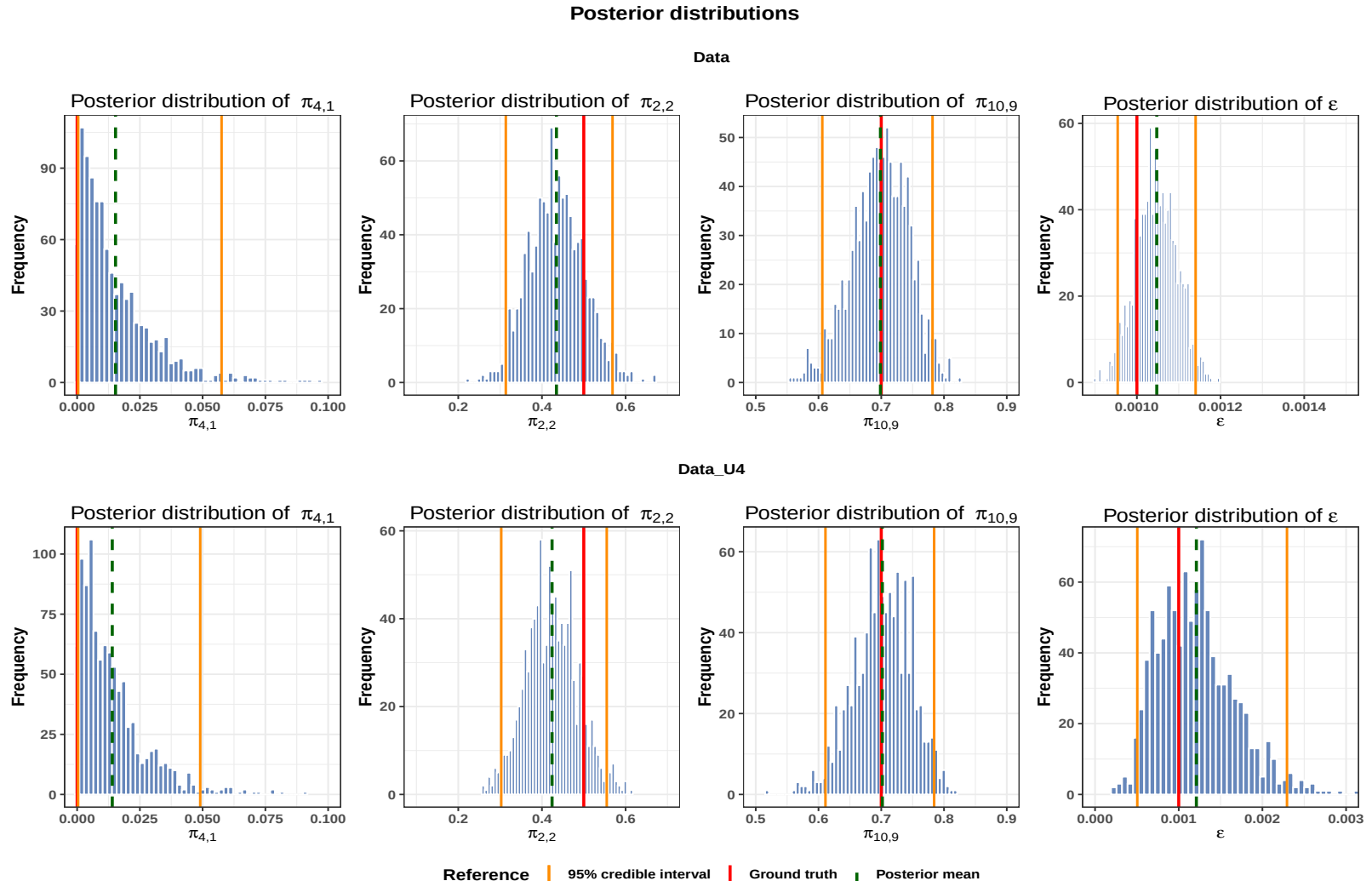
Best Model: Informative Position Subsampling + 3 Random Positions

Comparison of π Estimates



Best Model: Informative Position Subsampling + 3 Random Positions

Comparison of Posterior Distributions.



Thank you for your attention.

